

Initiating Search

November 10, 2025, 4:51 PM

• Search:

CAS Lexicon Search:

(Trifluoromethyl)trimethylsilane - Preferred Concept

(Trifluoromethyl)trimethylsilane - Broader Concepts (1)

Haloalkyl silanes

Filtered By:

Document Type: **Journal**Flags: **Has Related Reactions**Substance Role: **Reactant or Reagent**

Search Tasks

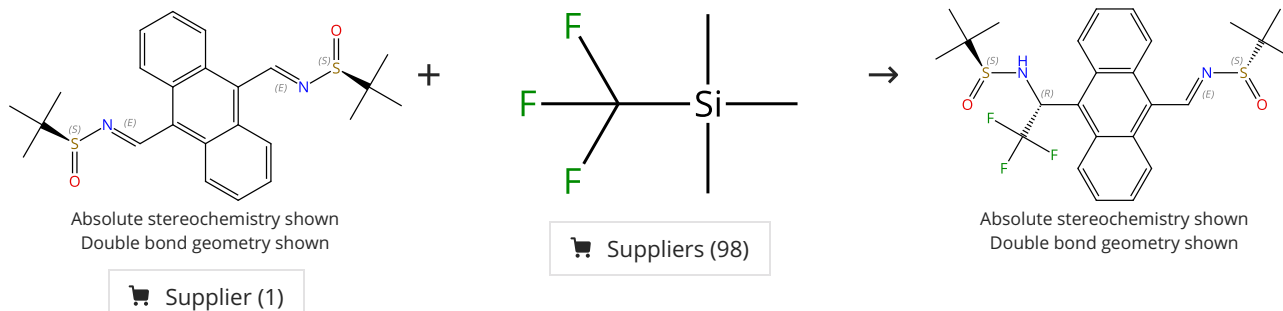
Task	Result Type	View
Returned Reference Results from CAS Lexicon + Filters (2,027)	 References	View Results
Exported: Retrieved Related Reaction Results + Filters (1,277)	 Reactions	View Results
Filtered By:		
Yield: 90-100%, 80-89%, 70-79%		

Reactions (50)

[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%



31-345-CAS-7489606

Steps: 1 Yield: 100%

Preparation of (1R,1'R)-1,1'-(anthracene-9,10-diyl)bis(2,2,2-trifluoroethanamine): a chiral diamine with low basicity

By: Estivill, Carla; et al

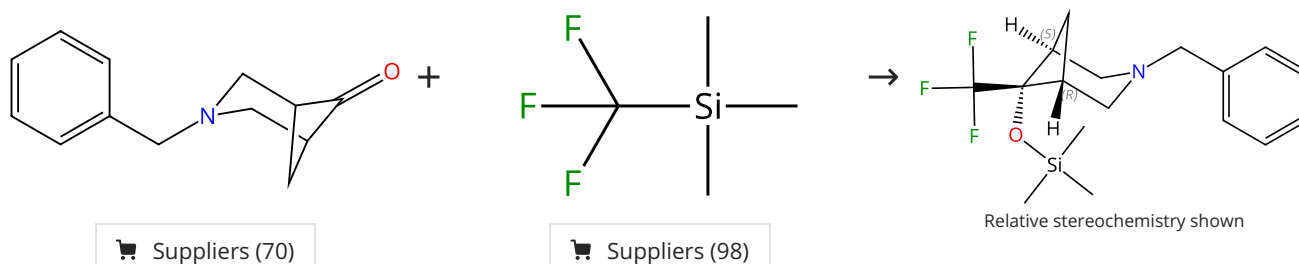
Tetrahedron (2009), 65(1), 171-176.

- 1.1 **Reagents:** Tetrabutylammonium difluorotriphenylsilicate
Solvents: Tetrahydrofuran; rt
- 1.2 **Solvents:** Tetrahydrofuran; 4 h, 195 K
- 1.3 **Reagents:** Ammonium chloride; 195 K; 195 K → rt

Experimental Protocols

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%



31-345-CAS-2309685

Steps: 1 Yield: 100%

3-Benzyl-3-azabicyclo[3.1.1]heptan-6-one: A Promising Building Block for Medicinal Chemistry

By: Denisenko, Aleksandr V.; et al

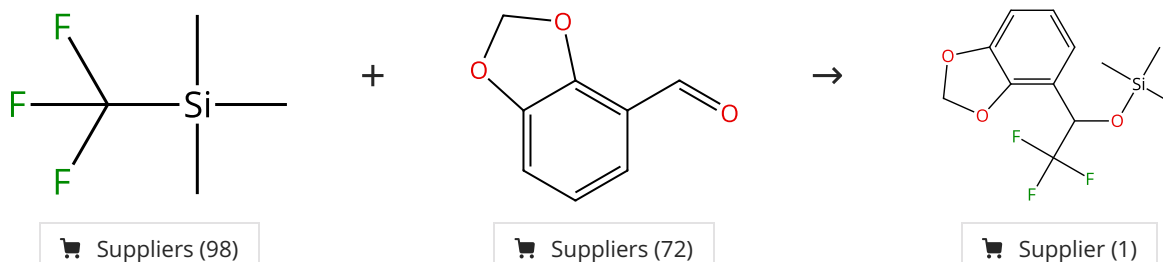
Organic Letters (2010), 12(19), 4372-4375.

- 1.1 **Reagents:** Cesium fluoride
Solvents: Tetrahydrofuran; -5 °C; 16 h, rt

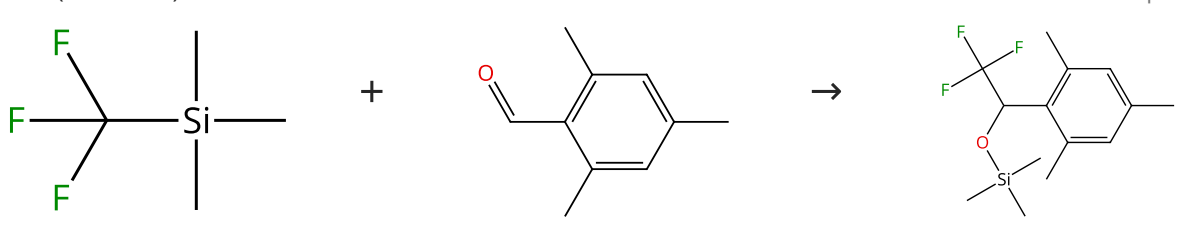
Experimental Protocols

Scheme 3 (1 Reaction)

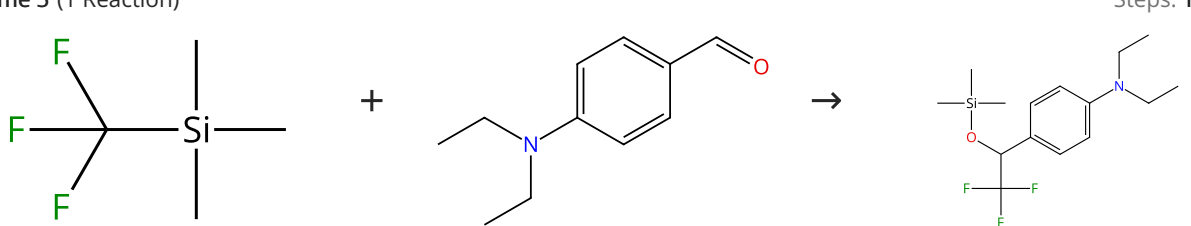
Steps: 1 Yield: 100%



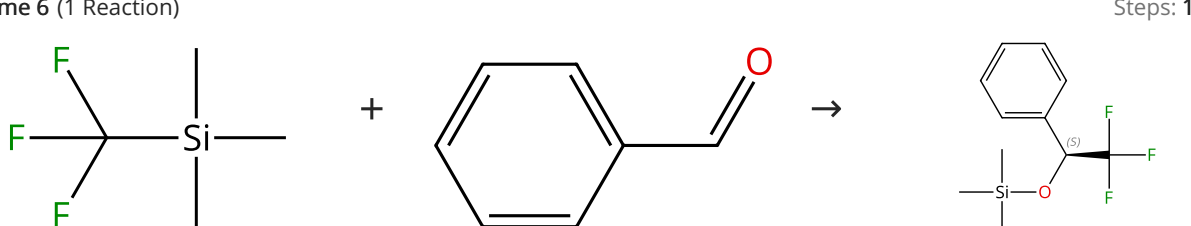
31-345-CAS-9680804 Steps: 1 Yield: 100% 1.1 Catalysts: Tetrabutylammonium fluoride Solvents: Tetrahydrofuran; 10 min, 10 °C; 5 h, 25 °C	Synthesis and characterization of novel trifluoromethyl-containing alcohols with Ruppert's reagent By: Singh, Rajendra P.; et al Journal of Fluorine Chemistry (2012), 133, 20-26.
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Scheme 4 (1 Reaction) Steps: 1 Yield: 100%   Suppliers (98)  Suppliers (88)  Suppliers (2)	
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31-345-CAS-7754753 Steps: 1 Yield: 100% 1.1 Solvents: Dimethyl sulfoxide; 15 min, rt 1.2 pH 7, rt Experimental Protocols	A new and efficient method for the trifluoromethylation of carbonyl compounds with trifluoromethyltrimethylsilane in DMSO By: Iwanami, Katsuyuki; et al Synlett (2006), (1), 112-114.
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Scheme 5 (1 Reaction) Steps: 1 Yield: 100%   Suppliers (98)  Suppliers (87)  Suppliers (2)	
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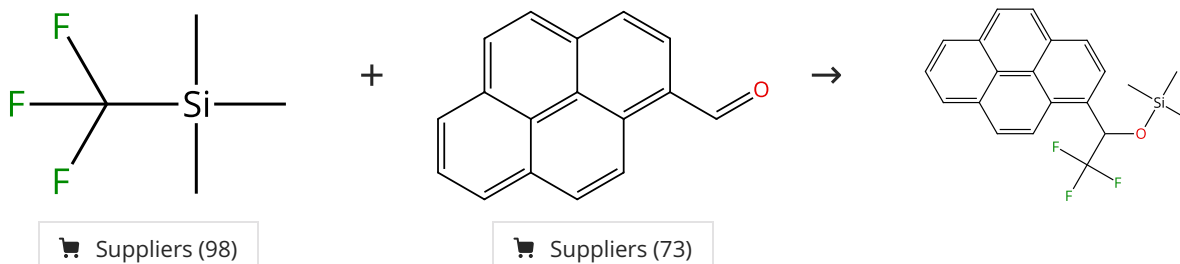
31-345-CAS-888408 Steps: 1 Yield: 100% 1.1 Reagents: Cesium fluoride Solvents: 1,2-Dimethoxyethane	Nucleophilic trifluoromethylation and difluorination of substituted aromatic aldehydes with Ruppert's and Deoxofluor reagents By: Singh, R. P.; et al Journal of Fluorine Chemistry (2001), 111(2), 153-160.
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Scheme 6 (1 Reaction) Steps: 1 Yield: 100%   Suppliers (98)  Suppliers (69) Absolute stereochemistry shown	
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31-345-CAS-4648988 Steps: 1 Yield: 100% 1.1 Catalysts: Potassium fluoride, (4<i>R</i>,4'<i>R</i>,5<i>R</i>,5'<i>R</i>)-2,2'-(1,4-Phenylene)bis[α,α',α'-tetraphenyl-1,3-dioxolane-4,5-dimethanol] Solvents: Acetonitrile; 1 h, 20 °C 1.2 Reagents: Water Solvents: Water; 20 °C 1.3 Reagents: Hydrofluoric acid Solvents: Acetonitrile, Water; 20 °C Experimental Protocols	Bis-TADDOLs as bifunctional organocatalysts for trifluoromethylation of aldehydes and ring-opening of cyclohexene oxide with <i>p</i>-toluidine By: Belokon, Yuri N.; et al Tetrahedron: Asymmetry (2010), 21(13-14), 1793-1796.
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Scheme 7 (1 Reaction)

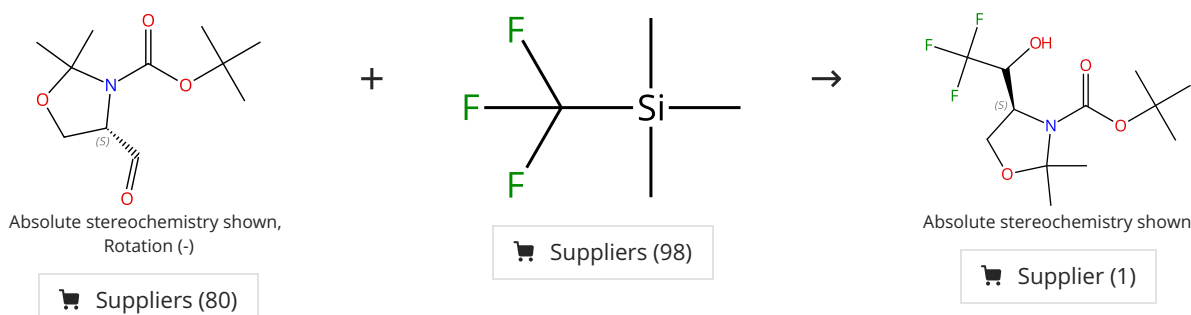
Steps: 1 Yield: 100%



31-345-CAS-5423397 Steps: 1 Yield: 100% 1.1 Catalysts: Tetrabutylammonium fluoride Solvents: Tetrahydrofuran; 10 min, 10 °C; 24 h, 25 °C	Synthesis and characterization of novel trifluoromethyl-containing alcohols with Ruppert's reagent By: Singh, Rajendra P.; et al Journal of Fluorine Chemistry (2012), 133, 20-26.
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Scheme 8 (1 Reaction)

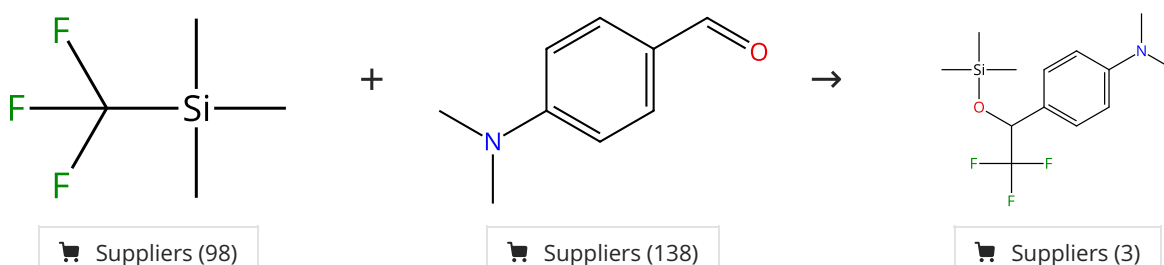
Steps: 1 Yield: 100%



31-345-CAS-9960909 Steps: 1 Yield: 100% 1.1 Reagents: Amberlyst 27 Solvents: Tetrahydrofuran 1.2 Reagents: Amberlyst 15 1.3 Reagents: AM (resin)	Synthesis of trifluoromethyl ketones using polymer-supported reagents By: Baxendale, I. R.; et al Combinatorial Chemistry and High Throughput Screening (2002), 5(3), 197-199.
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Scheme 9 (1 Reaction)

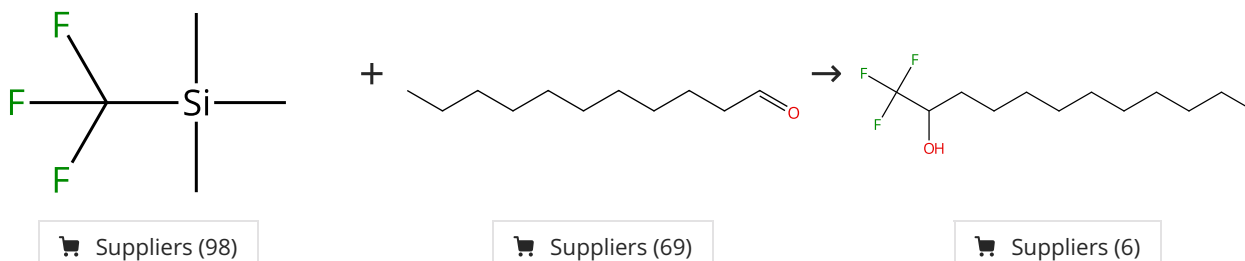
Steps: 1 Yield: 100%



31-345-CAS-12150425	Steps: 1 Yield: 100%	Nucleophilic trifluoromethylation and difluorination of substituted aromatic aldehydes with Ruppert's and Deoxofluor reagents
1.1 Reagents: Cesium fluoride Solvents: 1,2-Dimethoxyethane		By: Singh, R. P.; et al Journal of Fluorine Chemistry (2001), 111(2), 153-160.

Scheme 10 (1 Reaction)

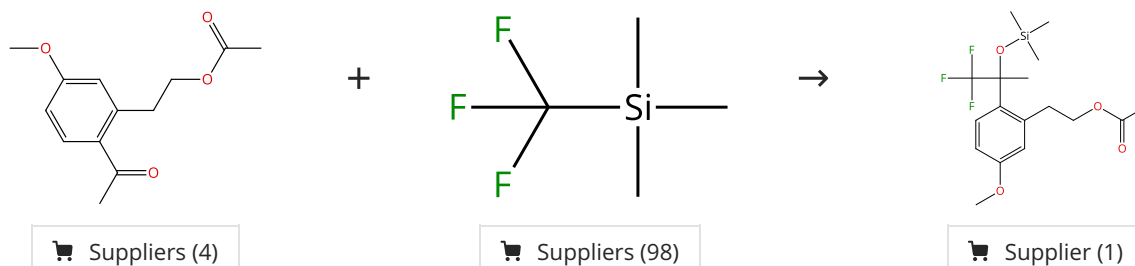
Steps: 1 Yield: 100%



31-345-CAS-12091111	Steps: 1 Yield: 100%	Synthesis of trifluoromethyl ketones using polymer-supported reagents
1.1 Reagents: Amberlyst 27 Solvents: Tetrahydrofuran		By: Baxendale, I. R.; et al
1.2 Reagents: Amberlyst 15		Combinatorial Chemistry and High Throughput Screening
1.3 Reagents: AM (resin)		(2002), 5(3), 197-199.

Scheme 11 (1 Reaction)

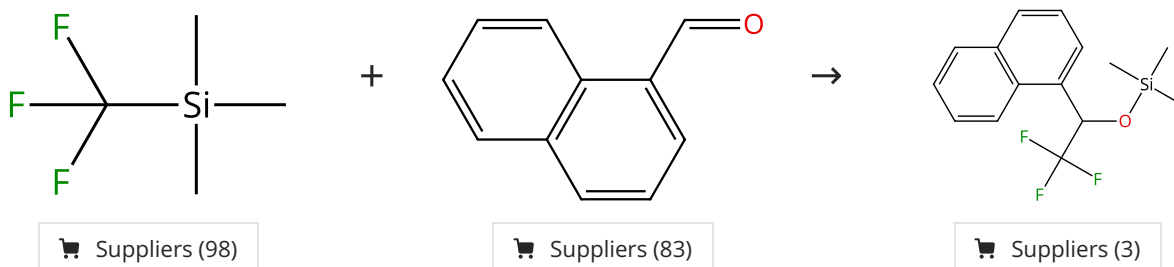
Steps: 1 Yield: 100%



31-345-CAS-11482333	Steps: 1 Yield: 100%	Enantioselective addition of a trifluoromethyl anion to aryl ketones and aldehydes
1.1 Reagents: Cesium fluoride Solvents: Dimethylformamide		By: Caron, Stephane; et al
Experimental Protocols		Synthesis (2003), (11), 1693-1698.

Scheme 12 (2 Reactions)

Steps: 1 Yield: 100%

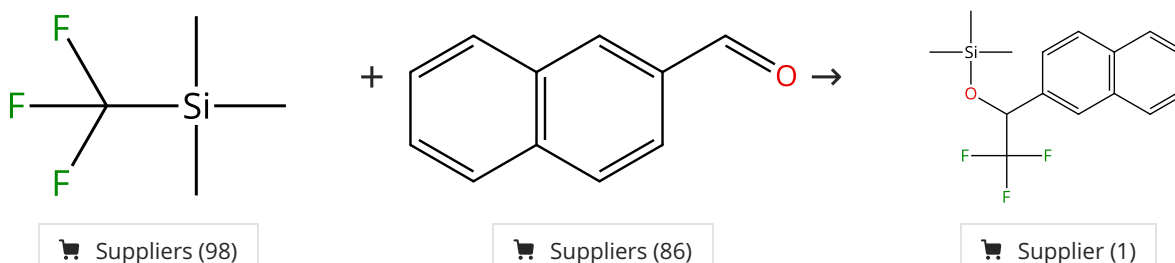


31-345-CAS-8899901 Steps: 1 Yield: 100% 1.1 Catalysts: Potassium fluoride, (4<i>R</i>,4'<i>R</i>,5<i>R</i>,5'<i>R</i>)-2,2'-(1,4-Phenylene)bis[α,α,α'-tetraphenyl-1,3-dioxolane-4,5-dimethanol] Solvents: Acetonitrile; 24 h, 20 °C 1.2 Reagents: Water Solvents: Water; 20 °C 1.3 Reagents: Hydrofluoric acid Solvents: Acetonitrile, Water; 20 °C Experimental Protocols	Bis-TADDOLs as bifunctional organocatalysts for trifluoromethylation of aldehydes and ring-opening of cyclohexene oxide with <i>p</i>-toluidine By: Belokon, Yuri N.; et al Tetrahedron: Asymmetry (2010), 21(13-14), 1793-1796.
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31-345-CAS-12285801 Steps: 1 Yield: 100% 1.1 Solvents: Dimethyl sulfoxide; 10 min, rt 1.2 pH 7, rt Experimental Protocols	A new and efficient method for the trifluoromethylation of carbonyl compounds with trifluoromethyltrimethylsilane in DMSO By: Iwanami, Katsuyuki; et al Synlett (2006), (1), 112-114.
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Scheme 13 (3 Reactions)

Steps: 1 Yield: 100%



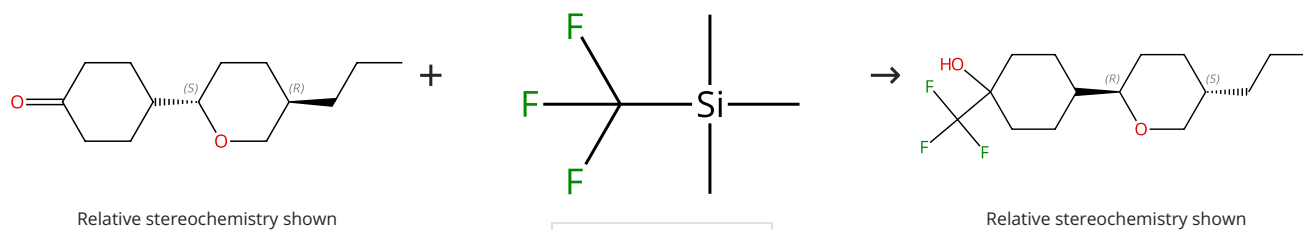
31-345-CAS-6777931 Steps: 1 Yield: 100% 1.1 Catalysts: Potassium fluoride, (4<i>R</i>,4'<i>R</i>,5<i>R</i>,5'<i>R</i>)-2,2'-(1,4-Phenylene)bis[α,α,α'-tetraphenyl-1,3-dioxolane-4,5-dimethanol] Solvents: Acetonitrile; 24 h, 20 °C 1.2 Reagents: Water Solvents: Water; 20 °C 1.3 Reagents: Hydrofluoric acid Solvents: Acetonitrile, Water; 20 °C Experimental Protocols	Bis-TADDOLs as bifunctional organocatalysts for trifluoromethylation of aldehydes and ring-opening of cyclohexene oxide with <i>p</i>-toluidine By: Belokon, Yuri N.; et al Tetrahedron: Asymmetry (2010), 21(13-14), 1793-1796.
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31-345-CAS-10457174 Steps: 1 Yield: 100% 1.1 Reagents: Potassium hydroxide Solvents: 1,1,1,3,3-Pentafluorobutane; 5 h, rt Experimental Protocols	Solkane 365mfc is an environmentally benign alternative solvent for trifluoromethylation reactions By: Kusuda, Akihiro; et al Green Chemistry (2009), 11(11), 1733-1735.
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31-345-CAS-14414061 Steps: 1 Yield: 100% 1.1 Solvents: Dimethyl sulfoxide; 10 min, rt 1.2 pH 7, rt Experimental Protocols	A new and efficient method for the trifluoromethylation of carbonyl compounds with trifluoromethyltrimethylsilane in DMSO By: Iwanami, Katsuyuki; et al Synlett (2006), (1), 112-114.
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Scheme 14 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (98)

31-345-CAS-12653838

Steps: 1 Yield: 100%

Super-fluorinated liquid crystals: towards the limits of polarity

1.1 Reagents: Tetrabutylammonium fluoride

Solvents: Tetrahydrofuran; 0 °C → rt

By: Kirsch, Peer; et al

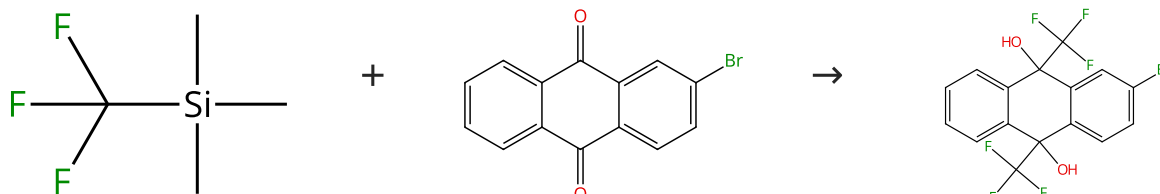
1.2 Reagents: Potassium fluoride

Solvents: Methanol; 3 h, reflux

European Journal of Organic Chemistry (2008), (20), 3479-3487.

Scheme 15 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (98)

Suppliers (74)

31-345-CAS-8788170

Steps: 1 Yield: 100%

Development of novel synthetic routes to bis(perfluoroalkyl)-substituted anthracene derivatives

1.1 Reagents: Potassium carbonate

Solvents: Dimethylformamide; 0 °C; 15 h, rt

By: Yamada, Shigeyuki; et al

1.2 Reagents: Hydrochloric acid

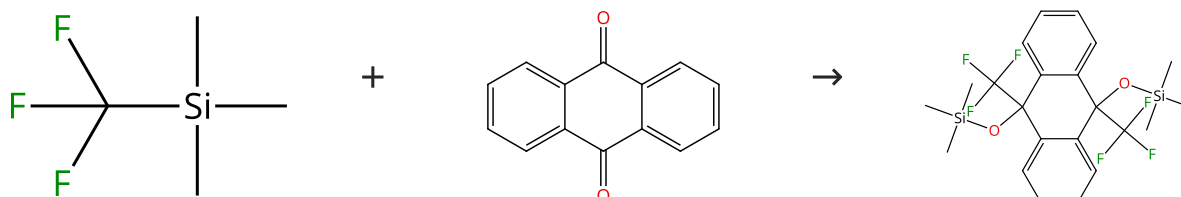
Solvents: Ethanol, Water; 3 h, reflux

RSC Advances (2013), 3(19), 6803-6806.

Experimental Protocols

Scheme 16 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (98)

Suppliers (105)

31-345-CAS-3288249

Steps: 1 Yield: 100%

Synthesis and characterization of novel trifluoromethyl-containing alcohols with Ruppert's reagent

1.1 Catalysts: Tetrabutylammonium fluoride

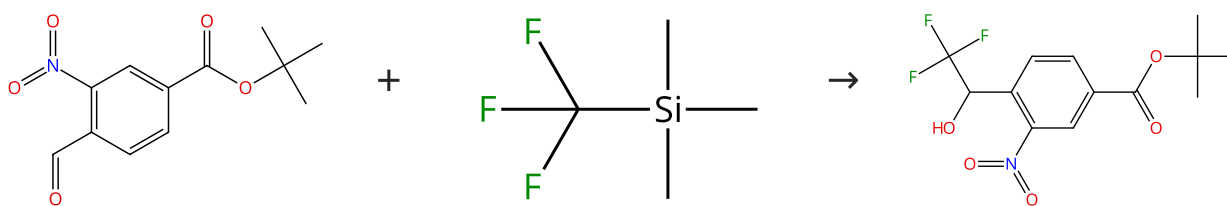
Solvents: Tetrahydrofuran; 10 min, 10 °C; 24 h, 25 °C

By: Singh, Rajendra P.; et al

Journal of Fluorine Chemistry (2012), 133, 20-26.

Scheme 17 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (5)

Suppliers (98)

31-345-CAS-12946495

Steps: 1 Yield: 100%

Selective Colorimetric Sensing of Anions in Aqueous Media through Reversible Covalent Bonding

By: Kim, Dae-Sik; et al

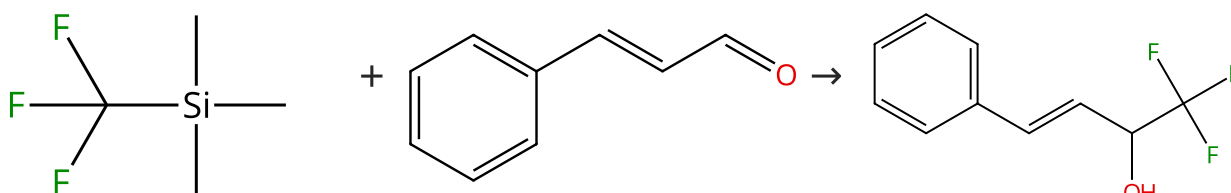
Journal of Organic Chemistry (2009), 74(13), 4849-4854.

- 1.1 Reagents: Cesium fluoride
Solvents: 1,2-Dimethoxyethane; 0 °C; 5 h, rt
- 1.2 Reagents: Hydrochloric acid
Solvents: Water; rt

Experimental Protocols

Scheme 18 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (98)

Suppliers (52)

Suppliers (9)

31-345-CAS-10283475

Steps: 1 Yield: 100%

Synthesis of trifluoromethyl ketones using polymer-supported reagents

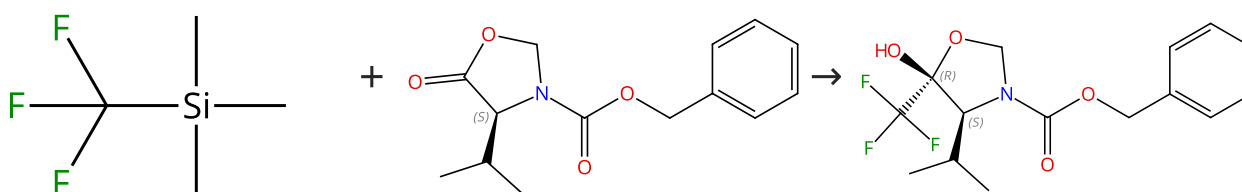
By: Baxendale, I. R.; et al

Combinatorial Chemistry and High Throughput Screening (2002), 5(3), 197-199.

- 1.1 Reagents: Amberlyst 27
Solvents: Tetrahydrofuran
- 1.2 Reagents: Amberlyst 15
- 1.3 Reagents: AM (resin)

Scheme 19 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (98)

Absolute stereochemistry shown,
Rotation (+)Absolute stereochemistry shown,
Rotation (+)

Suppliers (6)

Suppliers (2)

31-345-CAS-9671488

Steps: 1 Yield: 100%

Stereoselective synthesis of chiral β -amino trifluoromethyl alcohol: development of a manufacturing process for a key intermediate in the production of a novel elastase inhibitor, A E-3763

By: Inoue, Yasunao; et al

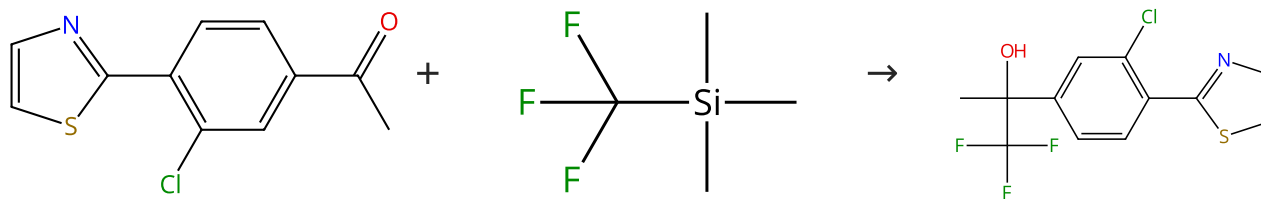
Tetrahedron: Asymmetry (2010), 21(15), 1855-1860.

- 1.1 Reagents: Cesium fluoride
Solvents: Tetrahydrofuran; rt $\rightarrow$ 0 °C; 30 min, 0 - 10 °C

Experimental Protocols

Scheme 20 (1 Reaction)

Steps: 1 Yield: 100%



Supplier (1)

Suppliers (98)

Supplier (1)

31-345-CAS-18193168

Steps: 1 Yield: 100%

- 1.1 **Catalysts:** Cesium fluoride
Solvents: Dimethylformamide; rt → 0 °C
- 1.2 1 h, 0 °C; 30 min, rt
- 1.3 **Reagents:** Tetrabutylammonium fluoride
Solvents: Tetrahydrofuran; 20 min, 0 °C; 0 °C → rt

Experimental Protocols

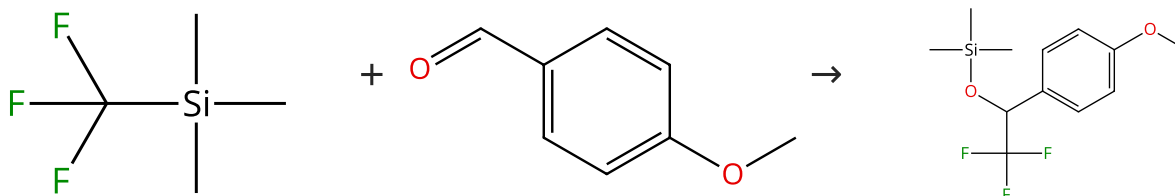
Small Molecule Disruptors of the Glucokinase-Glucokinase Regulatory Protein Interaction: 5. A Novel Aryl Sulfone Series, Optimization Through Conformational Analysis

By: Tamayo, Nuria A.; et al

Journal of Medicinal Chemistry (2015), 58(11), 4462-4482.

Scheme 21 (4 Reactions)

Steps: 1 Yield: 99-100%



Suppliers (98)

Suppliers (128)

Suppliers (3)

31-345-CAS-9878838

Steps: 1 Yield: 100%

- 1.1 **Solvents:** Dimethyl sulfoxide; 15 min, rt
- 1.2 pH 7, rt

Experimental Protocols

A new and efficient method for the trifluoromethylation of carbonyl compounds with trifluoromethyltrimethylsilane in DMSO

By: Iwanami, Katsuyuki; et al

Synlett (2006), (1), 112-114.

31-345-CAS-9681311

Steps: 1 Yield: 100%

- 1.1 **Reagents:** Cesium fluoride
Solvents: 1,2-Dimethoxyethane

Nucleophilic trifluoromethylation and difluorination of substituted aromatic aldehydes with Ruppert's and Deoxofluor reagents

By: Singh, R. P.; et al

Journal of Fluorine Chemistry (2001), 111(2), 153-160.

31-345-CAS-18241340

Steps: 1 Yield: 99%

- 1.1 **Catalysts:** 4-(Dimethylamino)pyridine, 1-Butanaminium, *N,N,N*-tributyl-, (*T*-4)-tricarboxylnitrosylferrate(1-) (1:1)
Solvents: Tetrahydrofuran; 14 h, 30 °C

Experimental Protocols

Fe-Catalyzed Nucleophilic Activation of C-Si versus Allylic C-O Bonds: Catalytic Trifluoromethylation of Carbonyl Groups versus Tandem Trifluoromethylation-Allylation of Olefins

By: Klein, Johannes E. M. N.; et al

Organometallics (2014), 33(20), 5802-5810.

31-345-CAS-334786

Steps: 1 Yield: 99%

- 1.1 **Catalysts:** Potassium acetate
Solvents: Dimethylformamide; 10 min, 0 °C

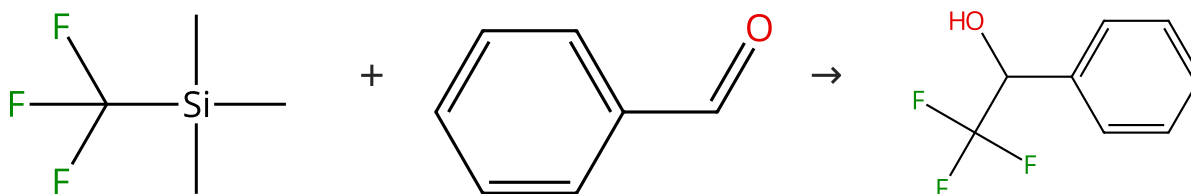
Lithium acetate-catalyzed trifluoromethylation of carbonyl compounds with (trifluoromethyl)trimethylsilane

By: Mukaiyama, Teruaki; et al

Chemistry Letters (2005), 34(1), 88-89.

Scheme 22 (3 Reactions)

Steps: 1 Yield: 99-100%



Suppliers (98)

Suppliers (69)

Suppliers (80)

31-345-CAS-2457685

Steps: 1 Yield: 100%

N-Heterocyclic Carbene Catalyzed Trifluoromethylation of Carbonyl Compounds

By: Song, Jinhua J.; et al

Organic Letters (2005), 7(11), 2193-2196.

1.1 **Catalysts:** 1,3-Bis[2,6-bis(1-methylethyl)phenyl]-1,3-dihydro-2*H*-imidazol-2-ylidene**Solvents:** Dimethylformamide; 0.5 h, rt1.2 **Reagents:** Hydrochloric acid**Solvents:** Water; rt

Experimental Protocols

31-345-CAS-10615215

Steps: 1 Yield: 100%

Synthesis of trifluoromethyl ketones using polymer-supported reagents

By: Baxendale, I. R.; et al

Combinatorial Chemistry and High Throughput Screening (2002), 5(3), 197-199.

1.1 **Reagents:** Amberlyst 27**Solvents:** Tetrahydrofuran1.2 **Reagents:** Amberlyst 151.3 **Reagents:** AM (resin)

31-345-CAS-14076008

Steps: 1 Yield: 99%

Highly efficient synthesis of aryl and heteroaryl trifluoromethyl ketones via o-iodoxybenzoic acid (IBX)

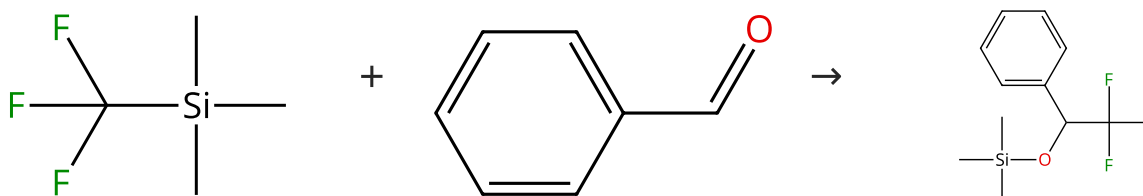
By: Cheng, Huicheng; et al

Tetrahedron Letters (2013), 54(33), 4483-4486.

1.1 **Catalysts:** Potassium carbonate**Solvents:** Dimethylformamide; 4 h, rt1.2 **Reagents:** Water; 4 h, rt

Scheme 23 (2 Reactions)

Steps: 1 Yield: 99-100%



Suppliers (98)

Suppliers (69)

Suppliers (3)

31-345-CAS-14747683

Steps: 1 Yield: 100%

A new and efficient method for the trifluoromethylation of carbonyl compounds with trifluoromethyltrimethylsilane in DMSO

By: Iwanami, Katsuyuki; et al

Synlett (2006), (1), 112-114.

1.1 **Solvents:** Dimethyl sulfoxide; 15 min, rt

1.2 pH 7, rt

Experimental Protocols

31-345-CAS-1929574

Steps: 1 Yield: 99%

Tri-*tert*-butylphosphine is an efficient promoter for the trifluoromethylation reactions of aldehydes, ketones, imides and imines

By: Mizuta, Satoshi; et al

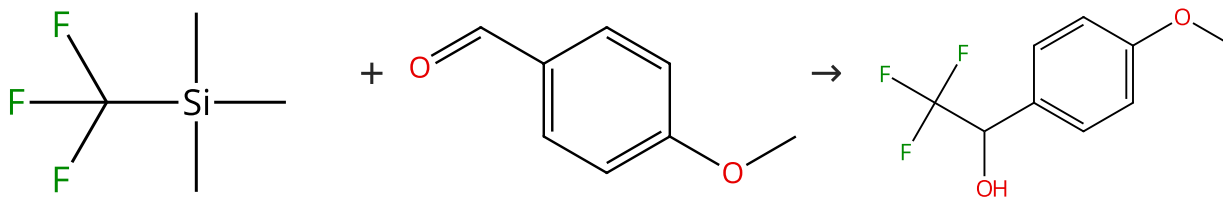
Synlett (2006), (2), 267-270.

1.1 **Catalysts:** Tri-*tert*-butylphosphine**Solvents:** Dimethylformamide; 15 - 60 min, rt

Experimental Protocols

Scheme 24 (2 Reactions)

Steps: 1 Yield: 99-100%



Suppliers (98)

Suppliers (128)

Suppliers (61)

31-345-CAS-3908908

Steps: 1 Yield: 100%

Synthesis of trifluoromethyl ketones using polymer-supported reagents

By: Baxendale, I. R.; et al

Combinatorial Chemistry and High Throughput Screening (2002), 5(3), 197-199.

- 1.1 Reagents: Amberlyst 27
Solvents: Tetrahydrofuran
- 1.2 Reagents: Amberlyst 15
- 1.3 Reagents: AM (resin)

31-345-CAS-6816781

Steps: 1 Yield: 99%

Highly efficient synthesis of aryl and heteroaryl trifluoromethyl ketones via o-iodoxybenzoic acid (IBX)

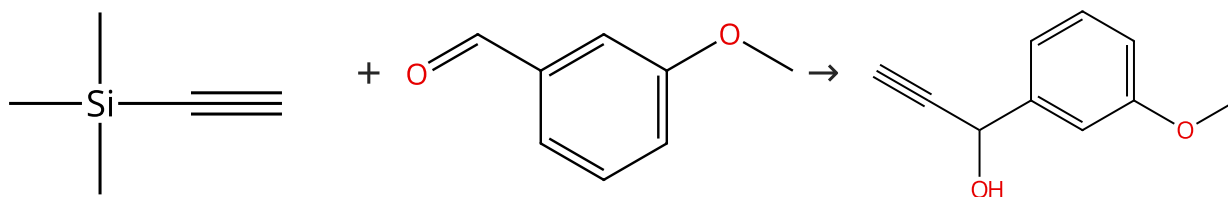
By: Cheng, Huicheng; et al

Tetrahedron Letters (2013), 54(33), 4483-4486.

- 1.1 Catalysts: Potassium carbonate
Solvents: Dimethylformamide; 4 h, rt
- 1.2 Reagents: Water; 4 h, rt

Scheme 25 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (98)

Suppliers (89)

Suppliers (21)

31-345-CAS-21122255

Steps: 1 Yield: 100%

A thiocyanopalladation/carbocyclization transformation identified through enzymatic screening: stereocontrolled tandem C-SCN and C-C bond formation

By: Malik, G.; et al

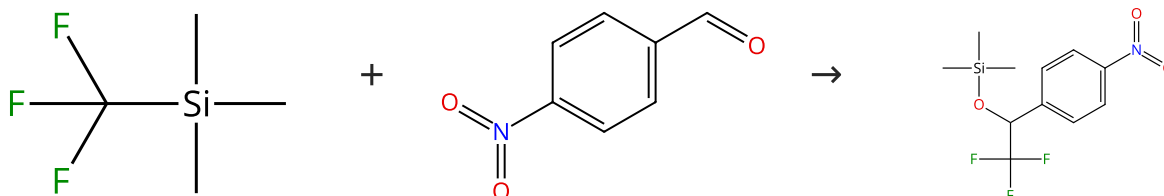
Chemical Science (2017), 8(12), 8050-8060.

- 1.1 Reagents: Butyllithium
Solvents: Tetrahydrofuran, Hexane; -78 °C; 30 min, -78 °C
- 1.2 -78 °C → 0 °C; 1 - 2 h, 0 °C
- 1.3 Reagents: Ammonium chloride
Solvents: Water
- 1.4 Reagents: Potassium carbonate
Solvents: Methanol; rt

Experimental Protocols

Scheme 26 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (98)

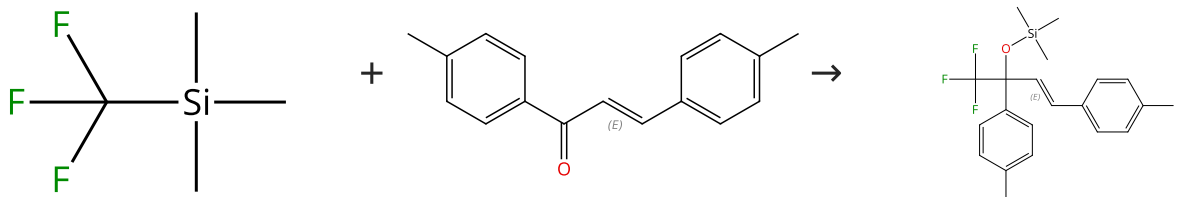
Suppliers (92)

Suppliers (5)

31-345-CAS-11036912 Steps: 1 Yield: 100% 1.1 Catalysts: Potassium fluoride, (4<i>R</i>,4'<i>R</i>,5<i>R</i>,5'<i>R</i>)-2,2'-(1,4-Phenylene)bis[α,α',α'-tetraphenyl-1,3-dioxolane-4,5-dimethanol] Solvents: Acetonitrile; 24 h, 20 °C 1.2 Reagents: Water Solvents: Water; 20 °C 1.3 Reagents: Hydrofluoric acid Solvents: Acetonitrile, Water; 20 °C Experimental Protocols	Bis-TADDOLs as bifunctional organocatalysts for trifluoromethylation of aldehydes and ring-opening of cyclohexene oxide with <i>p</i>-toluidine By: Belokon, Yuri N.; et al Tetrahedron: Asymmetry (2010), 21(13-14), 1793-1796.
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Scheme 27 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (98)

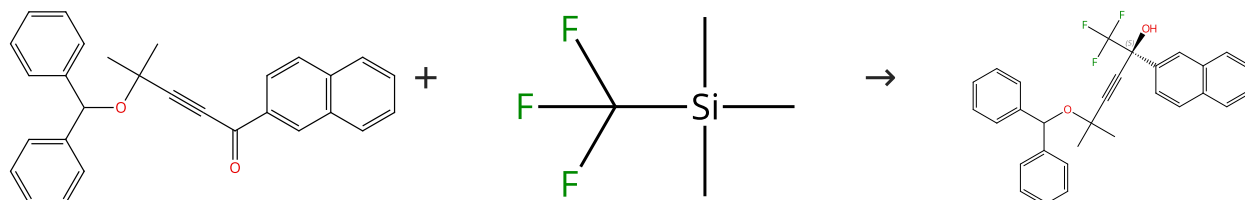
Double bond geometry shown

Suppliers (16)

31-345-CAS-21165805 Steps: 1 Yield: 99% 1.1 Reagents: Cesium fluoride Solvents: Diethyl ether; 4 h, rt	Acid-promoted cyclization of 2,4-diaryl-1,1,1-trifluorobut-3-en-2-oles and their TMS-ethers into CF₃-indenes By: Martynov, Matvei Yu.; et al Organic & Biomolecular Chemistry (2017), 15(12), 2541-2550.
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Scheme 28 (1 Reaction)

Steps: 1 Yield: 99%



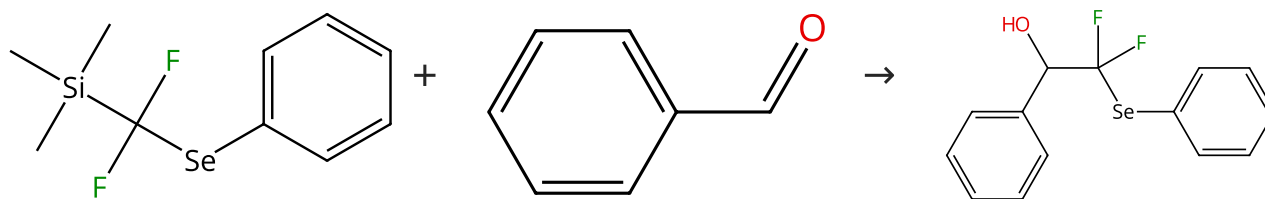
Suppliers (98)

Absolute stereochemistry shown,
Rotation (+)

31-345-CAS-3866641 Steps: 1 Yield: 99% 1.1 Catalysts: Tetramethylammonium fluoride, Cinchonanium, 1, 1''-[1,3-phenylenebis(methylene)]bis[9-hydroxy-, bromide (1:2), (9<i>S</i>)-(9''<i>S</i>)- Solvents: Dichloromethane, Toluene; 2 h, -60 °C 1.2 Reagents: Ammonium chloride Solvents: Water 1.3 Reagents: Tetrabutylammonium fluoride Solvents: Tetrahydrofuran; 1 h, rt Experimental Protocols	Cinchona Alkaloid-Catalyzed Asymmetric Trifluoromethylation of Alkynyl Ketones with Trimethylsilyl Trifluoromethane By: Kawai, Hiroyuki; et al Organic Letters (2010), 12(22), 5104-5107.
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Scheme 29 (1 Reaction)

Steps: 1 Yield: 99%



Supplier (1)

Suppliers (69)

Suppliers (2)

31-345-CAS-7335938

Steps: 1 Yield: 99%

Lewis acid-catalyzed tri- and difluoromethylation reactions of aldehydes

By: Mizuta, Satoshi; et al

Chemical Communications (Cambridge, United Kingdom) (2006), (24), 2575-2577.

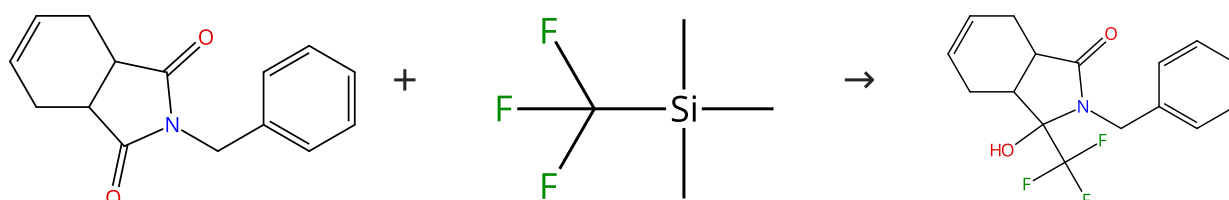
1.1 **Catalysts:** Cupric acetate, 1,2-Bis(diphenylphosphino)ethane
Solvents: Dimethylformamide; 30 min, rt

1.2 2 h, rt

1.3 **Reagents:** Hydrogen ion

Scheme 30 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (40)

Suppliers (98)

31-345-CAS-8792712

Steps: 1 Yield: 99%

Nucleophilic trifluoromethylation of cyclic imides using (trifluoromethyl)trimethylsilane CF₃SiMe₃

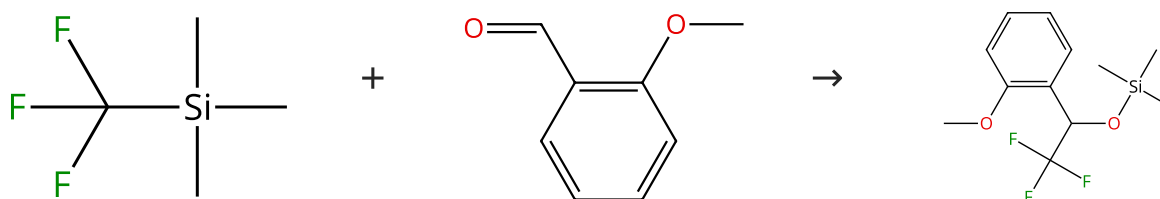
By: Hoffmann-Roeder, Anja; et al

Organic & Biomolecular Chemistry (2004), 2(16), 2267-2269.

1.1 **Reagents:** Cesium fluoride
Solvents: Tetrahydrofuran; 4 h, 0 °C → rt1.2 **Reagents:** Tetrabutylammonium fluoride
Solvents: Tetrahydrofuran; 30 min, rt1.3 **Reagents:** Hydrochloric acid
Solvents: Water; rt

Scheme 31 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (98)

Suppliers (96)

Suppliers (4)

31-345-CAS-5132958

Steps: 1 Yield: 99%

Nucleophilic trifluoromethylation and difluorination of substituted aromatic aldehydes with Ruppert's and Deoxofluor reagents

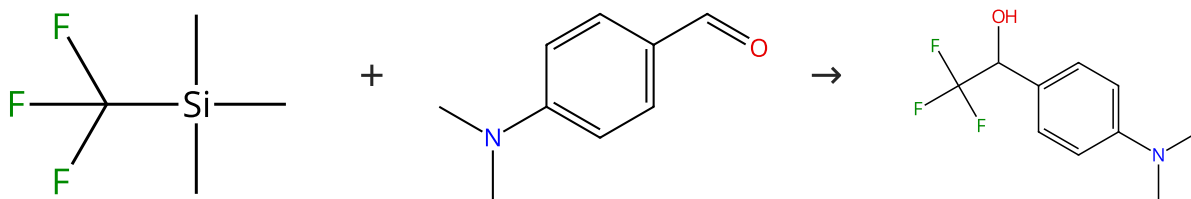
By: Singh, R. P.; et al

Journal of Fluorine Chemistry (2001), 111(2), 153-160.

1.1 **Reagents:** Cesium fluoride
Solvents: 1,2-Dimethoxyethane

Scheme 32 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (98)

Suppliers (138)

Suppliers (32)

31-345-CAS-4110129

Steps: 1 Yield: 99%

Highly efficient synthesis of aryl and heteroaryl trifluoromethyl ketones via o-iodoxybenzoic acid (IBX)

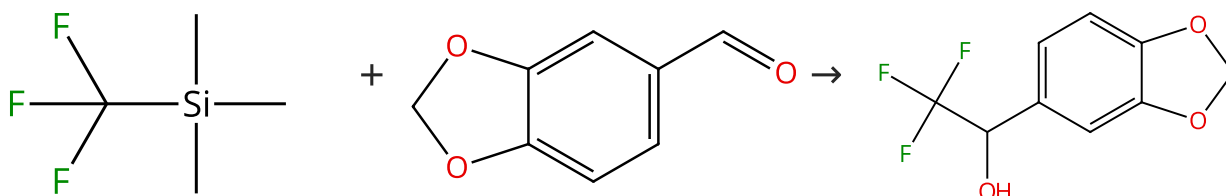
By: Cheng, Huicheng; et al

Tetrahedron Letters (2013), 54(33), 4483-4486.

- 1.1 **Catalysts:** Potassium carbonate
Solvents: Dimethylformamide; 4 h, rt
- 1.2 **Reagents:** Water; 4 h, rt

Scheme 33 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (98)

Suppliers (42)

Suppliers (44)

31-345-CAS-14926851

Steps: 1 Yield: 99%

Cinchona alkaloid/TMAF combination: Enantioselective trifluoromethylation of aryl aldehydes

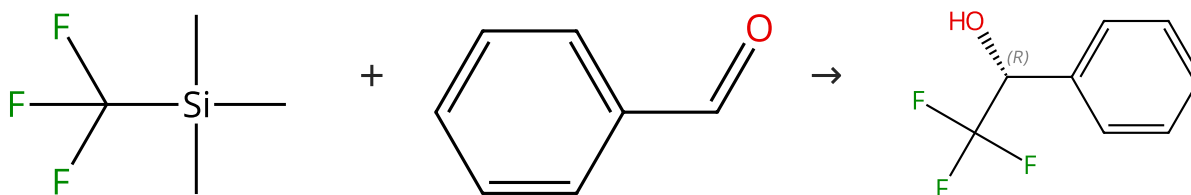
By: Kawai, Hiroyuki; et al

Journal of Fluorine Chemistry (2013), 152, 46-50.

- 1.1 **Catalysts:** Tetramethylammonium fluoride, Cinchonanium, 9-hydroxy-6'-methoxy-1-[[[3,3'',5,5''-tetrakis(trifluoromethyl)[1,1':3',1''-terphenyl]-5'-yl]methyl]-, bromide (1:1), (9S)-
Solvents: Dichloromethane, Toluene; 1 h, -60 °C
- 1.2 1 h, -60 °C
- 1.3 **Reagents:** Ammonium chloride
Solvents: Water
- 1.4 **Reagents:** Hydrochloric acid
Solvents: Water; 1 h, rt

Scheme 34 (2 Reactions)

Steps: 1 Yield: 99%



Suppliers (98)

Suppliers (69)

Suppliers (63)

31-345-CAS-11622666

Steps: 1 Yield: 99%

Asymmetric trifluoromethylation of aldehydes and ketones with trifluoromethyltrimethylsilane catalyzed by chiral quaternary ammonium fluorides

By: Iseki, Katsuhiko; et al

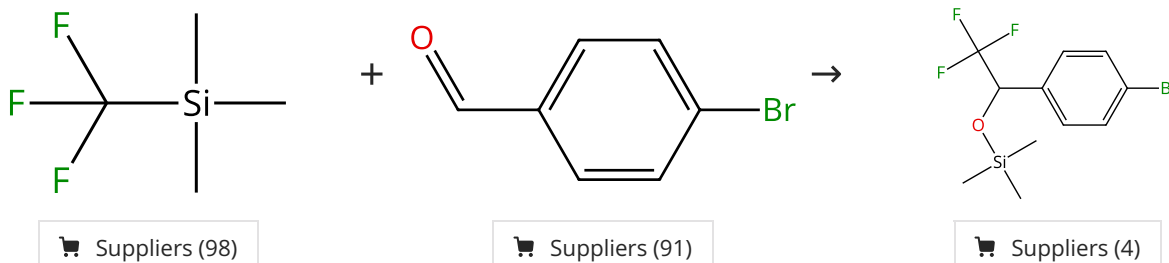
Tetrahedron Letters (1994), 35(19), 3137-8.

- 1.1 **Catalysts:** Cinchonanium, 9-hydroxy-1-[[4-(trifluoromethyl)phenyl]methyl]-, fluoride (1:1), (9S)-
Solvents: Toluene
- 1.2 **Reagents:** Hydrochloric acid
Solvents: Water

31-345-CAS-9495740	Steps: 1 Yield: 99%	Asymmetric trifluoromethylation of aldehydes and ketones with trifluoromethyltrimethylsilane catalyzed by chiral quaternary ammonium fluorides
1.1 Catalysts: Cinchonanium, 1-[[2,4-bis(trifluoromethyl)phenyl]methyl]-9-hydroxy-, fluoride, (9 <i>S</i>)- Solvents: Toluene		By: Iseki, Katsuhiko; et al
1.2 Reagents: Hydrochloric acid Solvents: Water		Tetrahedron Letters (1994), 35(19), 3137-8.

Scheme 35 (2 Reactions)

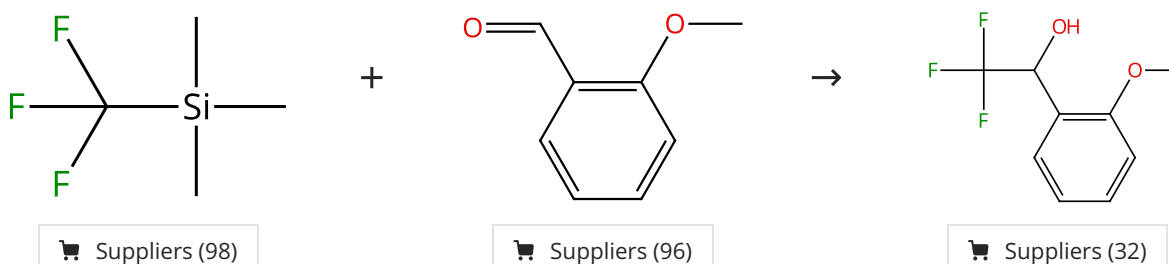
Steps: 1 Yield: 99%



31-345-CAS-18111696	Steps: 1 Yield: 99%	Fe-Catalyzed Nucleophilic Activation of C-Si versus Allylic C-O Bonds: Catalytic Trifluoromethylation of Carbonyl Groups versus Tandem Trifluoromethylation-Allylation of Olefins
1.1 Catalysts: 4-(Dimethylamino)pyridine, 1-Butanaminium, <i>N,N</i> , <i>N</i> -tributyl-, (7-4)-tricarbonylnitrosylferrate(1-) (1:1) Solvents: Tetrahydrofuran; 14 h, 30 °C		By: Klein, Johannes E. M. N.; et al
Experimental Protocols		Organometallics (2014), 33(20), 5802-5810.
31-345-CAS-11851607	Steps: 1 Yield: 99%	Ammonium bromides/KF catalyzed trifluoromethylation of carbonyl compounds with (trifluoromethyl)trimethylsilane and its application in the enantioselective trifluoromethylation reaction
1.1 Catalysts: Tetrabutylammonium bromide, Potassium fluoride Solvents: Tetrahydrofuran; 1 h, -20 °C		By: Mizuta, Satoshi; et al
1.2 Reagents: Hydrochloric acid Solvents: Water		Tetrahedron (2007), 63(35), 8521-8528.
Experimental Protocols		

Scheme 36 (1 Reaction)

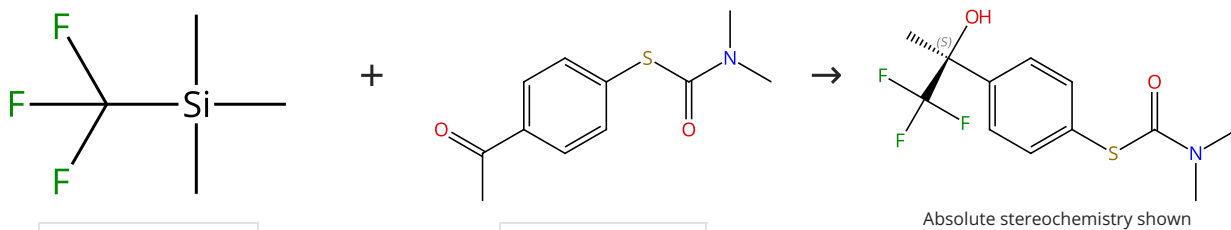
Steps: 1 Yield: 99%



31-345-CAS-11077699	Steps: 1 Yield: 99%	Highly efficient synthesis of aryl and heteroaryl trifluoromethyl ketones via o-iodoxybenzoic acid (IBX)
1.1 Catalysts: Potassium carbonate Solvents: Dimethylformamide; 4 h, rt		By: Cheng, Huicheng; et al
1.2 Reagents: Water; 4 h, rt		Tetrahedron Letters (2013), 54(33), 4483-4486.

Scheme 37 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (98)

Suppliers (4)

Absolute stereochemistry shown

31-345-CAS-18267237

Steps: 1 Yield: 99%

Synthesis, in Vitro Covalent Binding Evaluation, and Metabolism of ¹⁴C-Labeled Inhibitors of 11β-HSD1

By: Sun, Daqing; et al

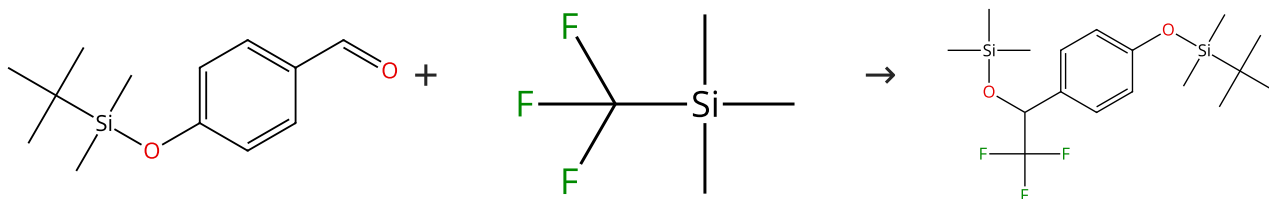
ACS Medicinal Chemistry Letters (2014), 5(11), 1245-1250.

- 1.1 **Reagents:** Tetrabutylammonium fluoride
Solvents: Tetrahydrofuran; -5 °C → 3 °C; 15 min, 6 °C; 20 min, 6 °C → 15 °C
- 1.2 **Reagents:** Hydrochloric acid
Solvents: Water; 15 °C → 40 °C; 30 min, 40 °C

Experimental Protocols

Scheme 38 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (65)

Suppliers (98)

31-345-CAS-2884143

Steps: 1 Yield: 99%

A new and efficient method for the trifluoromethylation of carbonyl compounds with trifluoromethyltrimethylsilane in DMSO

By: Iwanami, Katsuyuki; et al

Synlett (2006), (1), 112-114.

- 1.1 **Solvents:** Dimethyl sulfoxide; 15 min, rt
- 1.2 pH 7, rt

Experimental Protocols